Category-Level Band Learning-Based Feature Extraction for Hyperspectral Image Classification

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***Abstract*— Hyperspectral image (HSI) classification is a classical task in remote sensing image analysis. With the development of deep learning, schemes based on deep learning have gradually become the mainstream of HSI classification. However, existing HSI classification schemes either lack the exploration of category-specific information in the spectral bands and the intrinsic value of information contained in features at different scales, or are unable to extract multiscale spatial information and global spectral properties simultaneously. To solve these problems, in this article, we propose a novel HSI classification framework named CL-MGNet, which can fully exploit the category-specific properties in spectral bands and obtain features with multiscale spatial information and global spectral properties. Specifically, we first propose a spectral weight learning (SWL) module with a category consistency loss to achieve the enhancement of information in important bands and the mining of category-specific properties. Then, a multiscale backbone is proposed to extract the spatial information at different scales and the cross-channel attention via multiscale convolution and a grouping attention module. Finally, we employ an attention multilayer perceptron (attention-MLP) block to exploit the global spectral properties of HSI, which is helpful for the final fully connected layer to obtain the classification result. The experimental results on five representative hyperspectral remote sensing datasets demonstrate the superiority of our method.**

***Index Terms*— Attention mechanism, category-specific property, convolutional neural network (CNN), global property, hyperspectral image (HSI) classification, multiscale feature.**

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I. Introduction

HYPERSPECTRAL image (HSI) contains hundreds of continuous and narrow spectral bands with a high spectral resolution and is widely used in the field of remote sensing [1], [2], [3], [4], [5], [6]. HSI classification aims to assign a land-cover category to each pixel in HSI, which plays a crucial role in remote sensing image analysis (e.g., mineral exploration [7], agricultural planning [8], crop detection [9], and georeferencing inversion [10]), and has become a hot topic in the field of remote sensing in recent years [11], [12], [13]. On the one hand, the high spectral resolution of HSI can provide additional spectral information, which improves the accuracy of classification. On the other hand, the high dimensionality of HSI inevitably brings redundant information, causing the Hughes effect and making it difficult for traditional classification models to achieve satisfactory results [14], [15], [16], [17], [18].

To solve the problem of data redundancy caused by the high spectral resolution, a common approach is to use a series of statistical methods to extract key band information of HSI and remove other nonsignificant bands. Representative methods include principal component analysis (PCA) [19], factor analysis (FA) [20], and independent component analysis (ICA) [21]. Cao et al. [21] used ICA to reduce the dimensionality of HSI and obtain statistically independent features. Farrell and Mersereau [19] used PCA to perform a dimensionality reduction operation on real HSIs. Minhazur Rahman and Ahmed [20] eliminated the band correlation and retained useful information in a small number of spectral bands via FA. Although these methods can reduce the dimensionality of the HSI, they all use data correlation only as a basis for division and lack analysis of the spectral properties of HSI. As a result, the original physical and chemical properties of HSI are lost during dimensionality reduction.

Band selection is another common way to reduce redundant information, which aims to select a subset from the original band set with as little redundant information as possible [22], [23], [24], [25], [26], [27]. For example, Xu et al. [24] proposed a band selection method based on similarity ranking by using structural similarity (SSIM) to measure the relationship

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between bands. Wang et al. [25] treated band selection as a combinatorial optimization problem and achieved hyperspectral band selection by optimal neighborhood reconstruction (ONR). To minimize the loss of information, Fu et al. [27] proposed a band selection method based on a neighboring band grouping and matched filter ranking. Although these methods consider the spectral properties of HSI, most of them use only the linear relationship between spectral bands to select the important bands and lack consideration of the nonlinear relationship between spectral bands. Moreover, most of these methods tend to perform band selection at the dataset level, i.e., selecting a common subset of spectral bands for the whole dataset. However, due to the differences in the spectral reflectance characteristics of different categories, the important bands for each category are different, i.e., different band subsets should be selected for different categories. These dataset-level band selection methods fail to mine for category-specific information, resulting in bands containing this information not being selected correctly. This may lead to interference from redundant information and the omission of critical information in important bands.

In addition to the impact of redundant information on HSI classification, it is also a challenge to extract features containing more discriminative information from HSIs. Traditional HSI classification methods tend to use discrete cosine transform (DCT) [28], discrete wavelet transform (DWT) [29], linear discriminant analysis (LDA) [30], and nonparametric discriminant analysis (NDA) [31] to extract features from hyperspectral remote sensing images, and then combine with machine-learning-based classifiers (e.g., support vector machine (SVM) [32], k-nearest neighbor (kNN) [33], and Bayesian estimation [34]) to obtain the land-cover category of each pixel of remote sensing images. These methods can achieve good results when dealing with a small number of simple HSIs.

Recently, deep-learning-based models have shown stronger representation and generalization capabilities than the traditional machine-learning-based models [35], [36], [37], [38]. More and more researchers are working on the application of deep-learning-based models in HSI classification [39], [40], [41], [42]. For example, Zheng et al. [39] proposed a fully convolutional network (FCN)-based fast patch-free global learning framework, which can fuse the global spatial information and the semantic information of HSI to obtain the HSI classification map. Chakraborty and Trehan [40] combined the wavelet transform with the convolutional neural network (CNN) to extract additional spectral features and proposed a wavelet CNN for HSI classification. To exploit both spatial and spectral features in HSI, Zhong et al. [41] proposed a spectral–spatial residual network based on 3-D convolution to classify HSI. Similarly, Roy et al. [42] proposed an attention-based adaptive spectral–spatial kernel improved residual network (A2S2K-ResNet) to capture spectral–spatial features of HSI.

These CNN-based methods show powerful capability in extracting local features of HSI, but it has a shortcoming in capturing long-term dependencies. Recently, the transformer architecture has demonstrated strong performance in extracting long-term dependencies, spectral sequence properties, and high-level semantic features [43], [44], [45], [46], [47]. Sun et al. [44] proposed a spectral–spatial feature tokenization transformer (SSFTT) to capture both spectral–spatial features and high-level semantic features. Hong et al. [45] proposed a SpectralFormer for mining the sequence properties of spectral features to enhance the classification performance by learning the spectrally local sequence information in HSI. Yang et al. [46] achieved the caption and exploitation of spectral difference information and local spatial contextual information by combining convolutional operations with transformer. Similarly, He et al. [47] used a CNN to extract spatial features of HSI and used an improved transformer to capture sequential spectra relationships. Although these transformer-based methods have shown outstanding ability in extracting long-term spectral dependencies and spectral sequence properties, they have not well-exploited multiscale spatial features.

In HSIs, the spatial correlation of neighboring pixels is also a frequently used feature. However, the information of neighboring pixels under a single scale region is often limited to meet the needs of fine-grained classification. Considering that regions at different scales contain complementary information, it is necessary to capture the spatial information at different scales. Tu et al. [48] proposed a multiscale feature extraction (MSFE) method for HSI classification, using Gaussian pyramid decomposition to decompose the original HSI into a multiscale structure to capture the spatial information at different scales. Xue et al. [49] proposed a hierarchical residual network with an attention mechanism (HResNetAM) to extract features at different scales using a hierarchical approach. Moraga and Duzgun [50] extracted features at different scales using convolution kernels of different sizes and concated these features to obtain multiscale features. However, existing MSFE methods have primarily focused on researching manners for extracting multiscale spatial features, overlooking the intrinsic value of information contained in features at different scales. In fact, spatial features at different scales do not have the same benefits for classification, and features at some scales even have redundant information, which has negative effects on classification accuracy. In addition, the global spectral properties reflect the spectral differences between different pixels in HSI, which can also affect the accuracy of HSI classification.

In this article, we propose a novel HSI classification framework called CL-MGNet, which can fully exploit the category-specific properties in bands, multiscale spatial information, and global spectral properties of HSI. First, a spectral weight learning (SWL) module with a category consistency loss evaluates the weights of different bands through a spectral attention mechanism, which enhances the spectral properties of important bands and mines the category-specific spectral properties. Then, a new multiscale backbone is used to extract the multiscale spatial feature and the cross-channel attention, exploiting the valuable information in features at various scales while reducing the interference of redundant spatial features from less critical scales. At the same time, to take into account global spectral properties, we introduce an attention

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multilayer perceptron (attention-MLP) block to extract the global spectral properties of HSI to obtain the feature with more discriminative information. Finally, a fully connected layer is used to identify the obtained feature, assigning a land-cover category to each pixel of HSI. The experimental results on five representative hyperspectral remote sensing datasets demonstrate that our CL-MGNet has a better performance than the state-of-the-art frameworks. In summary, the main contributions of this article are as follows.

1) An attention-based module SWL with a category consistency loss is designed for mining important spectral information and category-specific properties in spectral bands.
2) To fully exploit the spatial correlation and contextual information between neighboring pixels in HSI, a multiscale backbone is proposed to extract the multiscale spatial feature with cross-channel attention.
3) An attention-MLP block is applied to extract the global spectral properties of HSI and obtain the feature embedding with more discriminative information.

The rest of this article is organized as follows. First, in Section II, we describe the details of the proposed CL-MGNet. Then, the experimental results and some analysis of this work are presented in Section III. Finally, we give a conclusion and discuss the future direction of this work in Section IV.

II. Proposed CL-MGNet

Hyperspectral remote sensing images provide abundant spectral information and bring a lot of redundant information that poses a challenge to the utilization of spectral properties and spatial contextual information [51], [52], [53]. There are some limitations of existing deep-learning-based HSI classification methods: 1) lacking the mining of category-specific spectral properties in bands and 2) the existing multiscale feature extractors overlook the intrinsic value of information contained in features at different scales and cannot well-extract multiscale spatial features and global spectral properties simultaneously. To address these limitations, we propose a CL-MGNet, which can simultaneously extract the category-specific properties in spectral bands, multiscale spatial information, and global properties of spectral signatures through three well-designed modules (i.e., an SWL module, a multiscale backbone, and an attention-MLP block).

In this section, we first provide a detailed introduction to each module of the proposed CL-MGNet. Then, we formulate the loss function for the CL-MGNet. Fig. 1 overviews the proposed CL-MGNet.

A. Category-Level SWL

Here, we propose a novel module named SWL, which can obtain the spectral weight of each category in HSI by simultaneously exploiting the linear and nonlinear relationships across spectral bands. Unlike the traditional spectral band attention module that assigns different band weights to samples belonging to the same category, the proposed SWL module uses a category consistency loss to ensure that band weights of samples from the same category are as consistent as possible, mining the category-specific properties in spectral bands.

Specifically, to measure the importance of the information contained in different bands of HSI, we use a spectral attention mechanism to calculate the weight of each band in HSI and use it to learn the spectral weight of each category in HSI. As shown at the top of Fig. 1, for an original HSI, we first divide it in the spatial dimension to obtain many different HSI patches. We use $\mathbf{H}_i$ to denote the ith HSI patch, where $\mathbf{H}_i \in \mathbb{R}^{B\times p\times p}$, B and p represent the number of spectral bands and the spatial size of the HSI patch, respectively. To calculate the spectral weight efficiently, we first aggregate the spatial information of the input HSI patch. Following [54], we use both the global max pooling (GMP) layer and the global average pooling (GAP) layer to generate two different spatial context descriptors: $\mathbf{D}_{\max} \in \mathbb{R}^{B\times 1\times 1}$ and $\mathbf{D}_{\text{avg}} \in \mathbb{R}^{B\times 1\times 1}$, i.e.,

$$\mathbf{D}_{\max} = \text{GMP}(\mathbf{H}_i) \quad \text{and} \quad \mathbf{D}_{\text{avg}} = \text{GAP}(\mathbf{H}_i). \tag{1}$$

These descriptors collect the max-pooled spectral features and the average-pooled spectral features of $\mathbf{H}_i$, respectively, which can be used to obtain finer spectral weight. Then, since these descriptors are located in the same semantic embedding space, we send them to a two-layer shared fully connected layer to infer the spectral weight. For convenience, we use two convolution layers with the kernel size of 1 to replace the two-layer shared fully connected layer, and use elementwise summation to merge the output feature vectors to obtain the spectral weight $\mathbf{w}_i \in \mathbb{R}^{B\times 1\times 1}$, i.e.,

$$\begin{aligned}\mathbf{w}_i = \sigma\big(&\mathbf{W}_2 * \text{ReLU}(\mathbf{W}_1 * \mathbf{D}_{\max}) \\ &+ \mathbf{W}_2 * \text{ReLU}\big(\mathbf{W}_1 * \mathbf{D}_{\text{avg}}\big)\big)\end{aligned} \tag{2}$$

where $*$ represents the convolution operation, and $\text{ReLU}(\cdot)$ is a nonlinear activation function. $\mathbf{W}_1$ and $\mathbf{W}_2$ represent the weights of the two convolution layers, respectively, where $\mathbf{W}_1 \in \mathbb{R}^{(B/\gamma)\times B}$, $\mathbf{W}_2 \in \mathbb{R}^{B\times(B/\gamma)}$, and γ is a reduction ratio used to reduce the computational overhead. In our experiments, we fix γ to 8. $\sigma(\cdot)$ represents the Sigmoid function to rescale the spectral weight between 0 and 1.

Thus, for the ith HSI patch $\mathbf{H}_i$, we can obtain its corresponding spectral weight $\mathbf{w}_i$, which indicates the importance of different spectral bands in $\mathbf{H}_i$. To further explore the category-specific spectral properties in different bands, we introduce a category consistency loss $\mathcal{L}_c$ to constrain the learning process of the SWL, and it is expressed as

$$\mathcal{L}_c = \frac{1}{N}\sum_{i=1}^{N} \|\mathbf{X}_{l_i} - \mathbf{w}_i\|_2^2 \tag{3}$$

where N denotes the number of input HSI patches, i.e., the number of samples, l_i represents the real land-cover category of ith sample, where $1 \le l_i \le M$, and M denotes the number of categories. $\mathbf{X}_{l_i}$ represents the spectral weight center of the l_ith category, which denotes the average of the spectral weights of the samples belonging to the l_ith category. In our experiments, $\mathbf{X}_{l_i}$ is a learnable parameter and it can be updated by model training.

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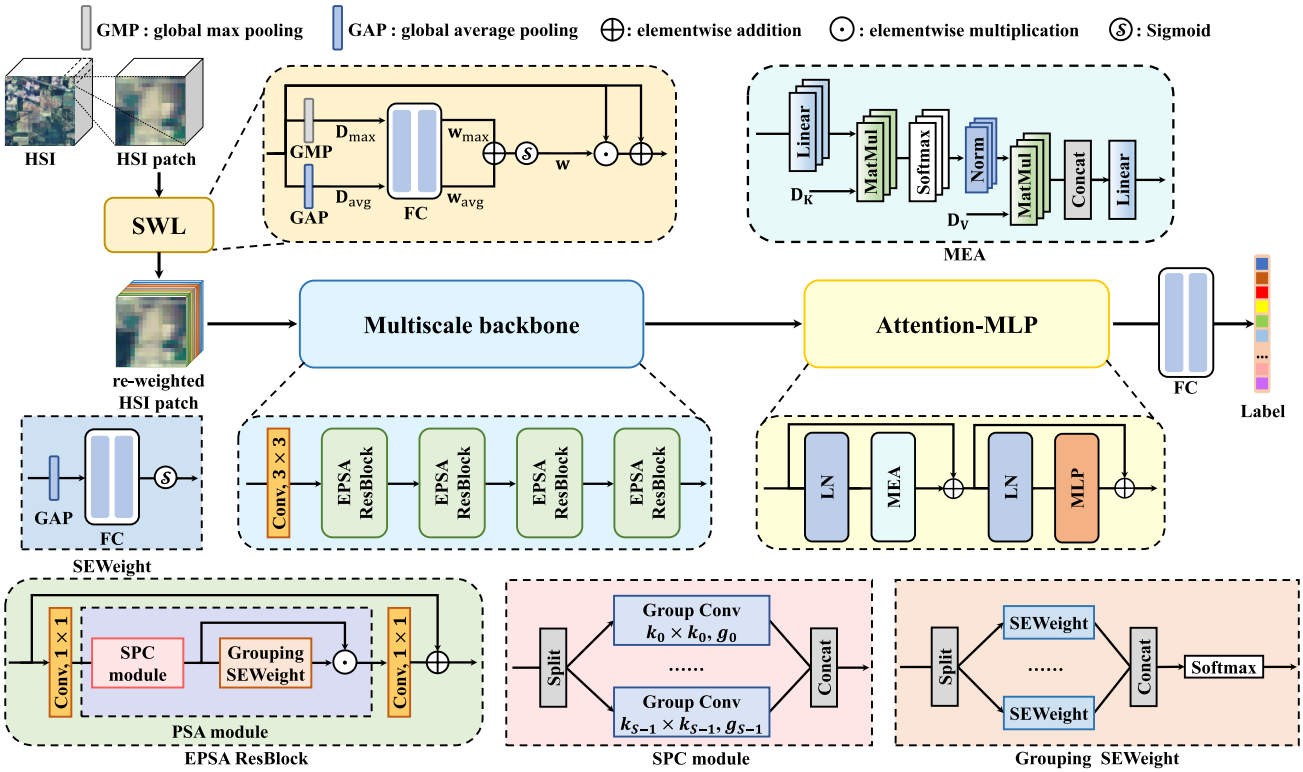


Fig. 1. Overview of the proposed CL-MGNet.

By introducing the category consistency loss to constrain the learning process of the SWL, the spectral weight gap between samples of the same category is reduced so that the spectral weights of samples of the same category obtained by the SWL are as close as possible to the spectral weight center of that category. Thus, the category-specific properties in spectral bands can be mined to the greatest extent, avoiding the omission of critical information. The reweighted HSI patch $\mathbf{H}_i'$ can be obtained as

$$\mathbf{H}_i' = \mathbf{H}_i \odot \mathbf{w}_i + \mathbf{H}_i \tag{4}$$

where $\odot$ denotes the elementwise multiplication.

B. Multiscale Spatial Feature With Cross-Channel Attention

To maximize the exploitation of the spatial information of HSI, we design a multiscale backbone to extract the multiscale spatial feature with cross-channel attention from the reweighted HSI patch $\mathbf{H}_i'$.

The overview of the multiscale backbone is shown in the middle of Fig. 1. For the reweighted HSI patch $\mathbf{H}_i'$, we first extract the shallow spatial feature through a 2-D convolution operation with the kernel size of 3. Then, the multiscale spatial information and the cross-channel attention are extracted by four efficient pyramid-squeeze attention (EPSA) ResBlocks [55]. As shown in the bottom left of Fig. 1, the first and third layers of the EPSA ResBlock are two convolutional layers with the kernel size of 1, which are used to ensure the consistency of channel dimensions. The second layer in the EPSA ResBlock is pyramid-squeezed attention (PSA) module, which can help extract the multiscale feature information with cross-channel attention. In addition, the skip connection is used to address the degradation problem.

In the PSA module, first, a multiscale spatial feature map is obtained from the input feature map in a multibranch manner by a squeeze pyramid concat (SPC) module. Specifically, for an input feature map $\mathbf{Z} \in \mathbb{R}^{C\times H\times W}$, it is first divided into S branches in the channel dimension, so the channels of the input feature map in each branch become $C' = (C/S)$. Then, in each branch, we employ group convolution [56] with different kernel sizes to obtain feature maps at different spatial receptive fields. However, as the convolution kernel size increases, the number of parameters of the network also increases drastically. To reduce the parameters of the model, a criterion for adaptively selecting the group size based on the kernel size of the group convolution is designed, and it can be described as

$$g = \begin{cases} 2^{\frac{k-1}{2}}, & k > 3 \\ 1, & 1 \le k \le 3 \end{cases} \tag{5}$$

where g denotes the group size of the group convolution, and k represents the kernel size of the group convolution.

In our experiments, the number of branches S is set to 4. The kernel size of the group convolution for each branch is set to 3, 5, 7, and 9 in turn (i.e., $k_0 = 3$, $k_1 = 5$, $k_2 = 7$, and $k_3 = 9$), and the group size of the group convolution for each branch can be obtained by (5), i.e., $g_0 = 1$, $g_1 = 4$, $g_2 = 8$, and $g_3 = 16$.

The SPC module is followed by a grouping SEWeight module, which is designed to extract the cross-channel attention from the multiscale spatial feature map. The cross-channel

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attention can adaptively regulate the weights of features at different scales to solve the negative impact of information at redundant scales on classification accuracy. As shown in the bottom right of Fig. 1, similar to the SPC module, we first divide the input multiscale spatial feature map into S branches. Then, we use an SEWeight module to obtain the channel attention of the input feature maps at different scales. SEWeight module is a commonly used channel attention module, which explicitly establishes the interdependence between feature channels and automatically obtains the importance of each channel through model training [57]. The structure of the SEweight module is shown in the middle left of Fig. 1. By using the SEWeight module, we can establish the cross-channel interaction in a local manner, obtaining the channel attention for inputs at different scales. Next, we concat the channel attention of different branches, and use a Softmax function to establish the interaction across all channels to obtain the cross-channel attention. Finally, the elementwise multiplication operation is applied to the cross-channel attention and the multiscale spatial feature map to obtain the multiscale spatial feature with cross-channel attention.

Through the multiscale backbone, we can fully exploit the spatial correlation and contextual information between neighboring pixels in HSI to obtain the finer multiscale spatial feature with cross-channel attention.

C. Global Spectral Properties' Extraction

To extract the global spectral properties of HSI, we introduce an attention-MLP block, as depicted in the middle right of Fig. 1. The attention-MLP block comprises a multihead external attention (MEA) mechanism [64], an MLP layer, and two normalization layers (layer normalization, LN). Additionally, skip connections are employed before the MEA mechanism and the MLP layer to enhance information flow.

The MEA mechanism is the core of the attention-MLP block, which can effectively capture the long-term dependencies of spectral signatures and extract global spectral properties. The details of the MEA mechanism are shown in the top right of Fig. 1. The key of the attention mechanism is to establish the dependencies between features and capture the global information from the entire feature space. However, the traditional self-attention (SA) mechanism only considers the relationship between elements in a single input when establishing the dependencies between features, ignoring the potential relationship between different inputs. To adequately capture the long-term dependencies among spectral features, the external attention (EA) mechanism replaces the key $\mathbf{K}$ and the value $\mathbf{V}$ in the traditional SA mechanism with two learnable memory units $\mathbf{D_K}$ and $\mathbf{D_V}$, respectively, introducing external samples to mine potential relationships among the whole dataset. The EA mechanism can be expressed as

$$\mathbf{F} = \mathrm{Norm}\left(\mathrm{Softmax}\left(\mathbf{Q}\mathbf{D_K}^{\mathrm{T}}\right)\right)\mathbf{D_V} \tag{6}$$

where $\mathbf{F}$ and $\mathbf{Q}$ represent the EA and the query matrix, respectively, $\mathrm{Norm}(\cdot)$ means the l_1-normalization, and $\mathrm{Softmax}(\cdot)$ denotes the Softmax function. By doing this, the EA mechanism no longer considers only the association of elements in a single input, but learns the potential knowledge from all inputs, improving its ability to establish the long-term dependencies among spectral signatures. The MEA mechanism involves the calculation of EA for multiple groups and can be formulated as

$$\hat{\mathbf{F}} = \mathrm{Concat}(\mathbf{F}_1, \mathbf{F}_2, \ldots, \mathbf{F}_h)\mathbf{A} \tag{7}$$

where $\hat{\mathbf{F}}$ denotes the MEA, h is the number of heads, and $\mathbf{A}$ is a linear transformation matrix used to ensure dimensionality consistency. In our method, h is set to 4.

Then, the MLP layer is used to further establish the mapping relationship between spectral features to obtain a more refined feature with global spectral properties. Through the attention-MLP module, the global spectral properties of the input multiscale spatial feature can be fully exploited and can make the obtained feature contain more discriminative information useful for classification. Finally, the obtained feature is sent to a fully connected layer to obtain the land-cover category of the input HSI patch.

D. Loss Function

Our CL-MGNet mainly contains three parts, i.e., SWL module, multiscale backbone, and attention-MLP block, and the output of the former part is fed directly to the latter part. To train these parts simultaneously, we introduce a cross-entropy loss function $\mathcal{L}_{\mathrm{ce}}$, which can be defined as

$$\mathcal{L}_{\mathrm{ce}} = -\frac{1}{N}\sum_{i=1}^{N}\log\left(p_{i,l_i}\right) \tag{8}$$

where p_{i,l_i} represents the prediction probability that the ith sample belongs to the l_ith category.

Thus, the total loss $\mathcal{L}$ of the CL-MGNet includes the category consistency loss for the SWL and the cross-entropy loss for classification, and is expressed as

$$\mathcal{L} = \mathcal{L}_{\mathrm{ce}} + \tau\mathcal{L}_c \tag{9}$$

where τ is a hyperparameter used to balance the category consistency loss and the cross-entropy loss. In the experiment, τ is set to 10 by trial and error.

III. Experimental Results

In this section, we first introduce five representative hyperspectral remote sensing datasets used in our experiments and some of our experimental settings. Then, we show the experimental results of the proposed CL-MGNet and compare it with the state-of-the-art HSI classification methods on five hyperspectral remote sensing datasets. Finally, we conduct experiments to verify the effectiveness of each module in our CL-MGNet.

A. Data Description

1) Indian Pines Dataset: This dataset is collected in north-western Indiana by the AVIRIS sensor and its spatial size is 145 × 145. The original Indian Pines dataset contains 224 spectral bands in the wavelength range of 0.4–2.5 μm and covers the region of water absorption. We use the corrected

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Indian Pines dataset after removing 24 spectral bands in the water absorption region for our experiments following [44]. The ground truth is designated into 16 categories.

2) Pavia University Dataset: This dataset is captured by the ROSIS sensor over Pavia, Nothern Italy. This scene has 103 spectral bands ranging from 430 to 860 nm with a spatial size of 610 × 340, and the geometric resolution is 1.3 m. The ground truth of Pavia University includes nine land-cover categories.

3) Houston2013 Dataset: This dataset is acquired over the University of Houston campus and the neighboring urban area [58]. This scene consists of 349 × 1905 pixels and has 144 spectral bands in the wavelength range of 380–1050 nm. The spatial resolution is 2.5 m. For classification purposes, 15 land-cover categories are defined for this dataset.

4) Dioni Dataset: The Dioni dataset is from the HyRANK benchmark [59], which is acquired by the Hyperion sensor on the National Aeronautics and Space Administration (NASA) Earth Observation 1 (EO-1) satellite. This dataset has 176 spectral bands and a spatial size of 250 × 1376. There are 12 land-cover categories in the Dioni dataset.

5) Houston2018 Dataset: This dataset is collected over the University of Houston campus using the ITRES CASI-1500 sensor, which is provided by the 2018 IEEE GRSS Data Fusion Competition [60]. The training portion of the full HSI is used for our experiments, and the ground truth is resampled to match the hyperspectral data. This dataset has a spatial size of 601 × 2384 and 48 spectral bands in the wavelength range of 380–1050 nm. There are 20 land-cover categories in the ground truth.

B. Experimental Setting

1) Evaluation Metrics: To measure the performance of the proposed method and other comparative methods, the commonly used classification evaluation metrics are selected for quantitative analysis, including overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa). The larger value of each metric indicates a better classification performance.

2) Sampling Strategy: To comprehensively evaluate the performance of the model, we incorporate two distinct sampling methods in our experiments. In the first situation, following [42] and [44], we randomly select training and test samples for each dataset. For the Indian Pines and Houston2013 datasets, we randomly select 15% samples for each category as the training set and the remaining samples as the test set. On the Pavia University, 5% samples for each category are randomly selected as the training set and the remaining samples are used as the test set. On the Dioni and Houston2018 datasets, we randomly select 10% samples for each category for training and the remaining samples for testing. In the second situation, we perform disjoint sampling strategy based on spatially disjoint training and test samples. For the Indian Pines, Pavia University, and Houston2013 datasets, these spatially disjoint training and test samples are available from the IEEE GRSS Data and Algorithm Standard Evaluation (DASE) website.[1]

[1]http://dase.grss-ieee.org

TABLE I
QUANTITATIVE CLASSIFICATION RESULTS USING DIFFERENT INPUT PATCH SIZES. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Dataset	Metrics	9×9	11×11	13×13	15×15
Indian Pines	OA(%)	99.51	**99.60**	99.37	99.51
	Kappa	0.9944	**0.9954**	0.9928	0.9944
Pavia University	OA(%)	99.69	99.75	**99.87**	99.72
	Kappa	0.9960	0.9967	**0.9983**	0.9962
Houston2013	OA(%)	99.70	99.84	**99.87**	99.85
	Kappa	0.9968	0.9983	**0.9986**	0.9984
Dioni	OA(%)	98.83	99.02	**99.32**	99.18
	Kappa	0.9855	0.9878	**0.9915**	0.9898
Houston2018	OA(%)	98.07	98.29	**98.40**	98.35
	Kappa	0.9746	0.9775	**0.9789**	0.9783

For the Dioni and Houston2018 datasets, we follow the disjoint sampling strategy and construct spatially disjoint training and test sets.

3) Experimental Configuration: Our experiments are run on an NVIDIA RTX 3090 GPU with the deep learning framework PyTorch. In the training phase, the batch size is fixed to 32, the stochastic gradient descent (SGD) [61] optimizer is used to optimize the learning process of the network, and the initial learning rate is set to 1×10^{-3} and finally decays to 1×10^{-5}. The training epoch for each dataset is set to 500.

C. Patch Size Analysis

Our method needs to separate the original HSI into HSI patches. The spatial size of the input HSI patch can affect the classification results. When the spatial size of the input HSI patch is large, it may cause the interference of irrelevant information, while a smaller patch size can make it difficult for the model to fully exploit the spatial correlation between neighboring pixels of the input HSI patch. To investigate the effect of patch size on the classification results, we experiment on five hyperspectral remote sensing datasets with patch sizes of 9 × 9, 11 × 11, 13 × 13, and 15 × 15, respectively, and the corresponding results are shown in Table I. It can be seen that the most suitable patch size for the Indian Pines dataset is 11 × 11. For the Pavia University, Houston2013, Dioni, and Houston2018 datasets, 13 × 13 is the most suitable patch size for them. In the experiments, we use these settings of patch size in our method.

D. Parameter Analysis

In the parameter analysis, we analyze several parameters that affect the classification performance and training process, including the number of branches of the PSA module S, the number of heads h, and the number of encoders of the attention-MLP. During the parameter analysis, other parameters (e.g., batch size, optimizer, epoches, and patch size) are configured as described in Sections III-B3 and III-C, and experiments are performed with randomly selected samples.

Table II shows the effect of the number of branches on classification accuracy of five datasets. Three different numbers of branches ($S = 2$, 4, and 8, respectively) are selected. It can be seen that when the number of branches is increased from 2 to 4, the OA on the Indian Pines, Pavia University,

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TABLE II
QUANTITATIVE RESULTS OF EXPERIMENTS WITH DIFFERENT NUMBER OF BRANCHES ON FIVE DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Number of Branches	Indian Pines		Pavia University		Houston2013		Dioni		Houston2018		Number of Parameters
	OA(%)	Kappa	OA(%)	Kappa	OA(%)	Kappa	OA(%)	Kappa	OA(%)	Kappa	
2	99.40	0.9932	99.77	0.9969	99.55	0.9951	99.19	0.9900	**98.41**	**0.9790**	**3.21M**
4	**99.60**	**0.9954**	**99.87**	**0.9983**	**99.87**	**0.9986**	**99.32**	**0.9915**	98.40	0.9789	3.82M
8	99.50	0.9943	99.81	0.9975	99.76	0.9974	99.23	0.9904	98.39	0.9788	4.59M

TABLE III
QUANTITATIVE RESULTS OF EXPERIMENTS WITH DIFFERENT NUMBER OF HEADS ON FIVE DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Number of Heads	Indian Pines		Pavia University		Houston2013		Dioni		Houston2018		Number of Parameters
	OA(%)	Kappa	OA(%)	Kappa	OA(%)	Kappa	OA(%)	Kappa	OA(%)	Kappa	
1	99.47	0.9940	99.83	0.9977	99.80	0.9979	99.24	0.9906	98.43	0.9793	3.843M
4	**99.60**	**0.9954**	**99.87**	**0.9983**	**99.87**	**0.9986**	**99.32**	**0.9915**	98.40	0.9789	3.818M
8	99.23	0.9912	99.69	0.9958	99.55	0.9952	99.18	0.9899	98.37	0.9785	3.814M
16	99.50	0.9943	99.80	0.9973	99.58	0.9954	99.23	0.9905	**98.46**	**0.9798**	**3.812M**

Houston2013, and Dioni datasets increases by 0.18%, 0.15%, 0.12%, and 0.05%, respectively, and the number of parameters is increased. However, when we further increase the number of branches to 8, the OA and Kappa of the model decreases. The reason may be that the redundant spatial contextual information resulting from the large scale of the extracted spatial features, leading to a reduction in the classification accuracy of the model.

Table III shows the impact of the number of heads for the MEA mechanism. We select four different numbers of heads (i.e., $h = 1$, 4, 8, and 16) to analyze the impact of the number of heads. As shown in Table III, on most datasets, the classification accuracy displays a trend of initially increasing and then decreasing. This trend arises from the unique perspective provided by each head in the multihead attention. A higher number of heads allows the model to explore the target from more angles. However, a larger number of heads does not guarantee improved model performance. On the contrary, some perspectives may introduce redundancy in the multihead attention, which can negatively impact the performance of the model.

Then, we explore the impact of the number of encoders for the attention-MLP block on classification results. As shown in Table IV, with the exception of the Houston2018 dataset, as the number of encoders increases, the OA and Kappa of the other four datasets exhibit a decline. This indicates that for most of the datasets, a single encoder is sufficient to achieve the best classification results since additional encoders may lead to network degradation instead of performance improvement.

E. Comparison With the State-of-the-Art Methods

To verify the effectiveness of the proposed CL-MGNet, several representative methods are selected for comparison, including HybridSN [62], A2S2K-ResNet [42], SpectralNET [40], attention-based second-order pooling network (A-SPN) [63], JigsawHSI [50], HResNetAM [49], SpectralFormer [45], and SSFTT [44]. All of these methods are deep-learning-based methods, where HybridSN, A2S2K-ResNet, JigsawHSI, and SpectralNET focus on extracting spectral–spatial features of HSI, and A-SPN aims to extract second-order statistical features. HResNetAM is a method to extract multiscale features. SpectralFormer and SSFTT use the transformer to extract sequence properties and high-level semantic information of HSI, respectively. For a fair comparison, we use the recommended parameter settings in their original article for all compared methods.

1) Classification Results With Randomly Selected Samples: In this experiment, we compare the classification performance of different methods based on randomly selected samples, i.e., 15% per category for the Indian Pines and Houston2013 datasets, 5% per category for the Pavia University dataset, and 10% per category for the Dioni and Houston2018 datasets. Tables V–IX show the experimental results of all methods on the Indian Pines, Pavia University, Houston2013, Dioni, and Houston2018 datasets, respectively.

In Table V, due to the serious sample imbalance problem on the Indian Pines dataset, both SpectralFormer and SSFTT suffer from overfitting in the small sample category, i.e., category 9, and only achieve the classification accuracy of 58.82%. However, our method fully exploits the category-specific properties in the spectral bands and combines the multiscale spatial feature with the global spectral properties to achieve the highest OA, AA, and Kappa of 99.60%, 99.71%, and 0.9954, respectively. In addition, on the Pavia University dataset, as shown in Table VI, our CL-MGNet effectively learns the important spectral information and extracts the feature with more discriminating information, achieving the highest OA, AA, and Kappa of 99.87%, 99.63%, and 0.9983, respectively. Compared with other methods, for example, the second highest result on the Pavia University dataset, A2S2K-ResNet achieves the OA of 99.65%, which is 0.22% away from our method in terms of OA. For the Houston2013

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TABLE IV

QUANTITATIVE RESULTS OF EXPERIMENTS WITH DIFFERENT NUMBER OF ENCODERS ON FIVE DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Number of Encoders	Indian Pines		Pavia University		Houston2013		Dioni		Houston2018		Number of Parameters
	OA(%)	Kappa	OA(%)	Kappa	OA(%)	Kappa	OA(%)	Kappa	OA(%)	Kappa	
1	**99.60**	**0.9954**	**99.87**	**0.9983**	**99.87**	**0.9986**	**99.32**	**0.9915**	98.40	0.9789	**3.82M**
2	99.44	0.9941	99.74	0.9965	99.80	0.9979	99.22	0.9904	98.41	0.9791	4.88M
3	99.47	0.9940	99.85	0.9980	99.77	0.9975	98.69	0.9838	98.41	0.9790	5.94M
4	99.12	0.9899	99.77	0.9970	99.71	0.9969	99.30	0.9913	**98.43**	**0.9793**	7.00M

TABLE V

QUANTITATIVE CLASSIFICATION RESULTS OF DIFFERENT METHODS FOR THE INDIAN PINES DATASET (WITH 15% TRAINING SAMPLES). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Category No.	HybridSN [62]	A2S2K-ResNet [42]	SpectralNET [40]	A-SPN [63]	JigsawHSI [50]	HResNetAM [49]	SpectralFormer [45]	SSFTT [44]	Ours
1	**100.00**	**100.00**	**100.00**	98.46	**100.00**	**100.00**	84.62	35.90	**100.00**
2	94.89	99.29	96.73	99.30	95.39	99.28	96.29	92.59	**99.34**
3	98.87	99.07	99.33	99.40	99.86	98.86	93.34	**99.72**	**99.72**
4	**100.00**	99.41	98.59	99.35	98.51	99.20	81.09	**100.00**	**100.00**
5	98.30	99.37	**99.77**	98.69	97.08	99.61	90.02	99.51	98.78
6	99.19	99.79	98.48	99.94	99.52	99.43	99.36	**100.00**	**100.00**
7	**100.00**	98.20	**100.00**	99.58	**100.00**	98.61	79.17	**100.00**	**100.00**
8	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
9	**100.00**	91.77	94.44	95.29	82.35	94.07	58.82	58.82	**100.00**
10	99.52	98.98	99.20	98.86	99.15	99.55	89.95	97.09	**99.76**
11	99.28	99.43	99.19	99.51	99.33	99.56	95.69	**99.62**	**99.62**
12	98.41	**99.16**	97.75	98.97	93.65	96.94	92.26	97.22	98.61
13	98.85	99.93	**100.00**	99.48	98.85	**100.00**	99.43	**100.00**	**100.00**
14	99.91	99.68	99.56	**99.93**	99.53	99.26	96.00	99.53	99.81
15	99.09	99.59	99.42	**99.91**	99.09	99.16	82.23	98.17	99.70
16	92.41	97.23	96.43	**100.00**	98.73	98.96	98.73	**100.00**	**100.00**
OA(%)	98.61	99.37	98.82	99.45	98.37	99.25	94.20	97.90	**99.60**
AA(%)	98.67	98.81	98.68	99.17	97.57	98.90	89.88	92.39	**99.71**
Kappa	0.9842	0.9928	0.9865	0.9937	0.9814	0.9914	0.9339	0.9760	**0.9954**

TABLE VI

QUANTITATIVE CLASSIFICATION RESULTS OF DIFFERENT METHODS FOR THE PAVIA UNIVERSITY DATASET (WITH 5% TRAINING SAMPLES). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Category No.	HybridSN [62]	A2S2K-ResNet [42]	SpectralNET [40]	A-SPN [63]	JigsawHSI [50]	HResNetAM [49]	SpectralFormer [45]	SSFTT [44]	Ours
1	99.94	99.97	98.98	99.70	99.43	99.26	97.08	99.94	**100.00**
2	99.97	99.98	99.94	**100.00**	99.99	99.81	99.54	99.86	99.98
3	99.65	99.51	95.54	95.39	95.89	97.61	87.96	99.40	**99.70**
4	98.35	**100.00**	98.97	99.04	99.35	99.58	98.01	98.21	99.35
5	98.28	99.67	99.84	**100.00**	98.75	99.80	**100.00**	98.51	99.84
6	**100.00**	99.63	99.79	99.33	**100.00**	99.71	99.69	**100.00**	**100.00**
7	99.84	**100.00**	97.47	**100.00**	98.97	99.24	89.23	**100.00**	**100.00**
8	97.17	97.10	97.48	99.20	97.48	95.22	87.39	99.46	**100.00**
9	93.44	**100.00**	94.67	99.89	97.11	99.33	99.89	97.78	97.78
OA(%)	99.39	99.65	99.08	99.51	99.29	99.14	97.15	99.63	**99.87**
AA(%)	98.52	99.54	98.08	99.17	98.52	98.84	95.42	99.24	**99.63**
Kappa	0.9920	0.9954	0.9878	0.9935	0.9906	0.9886	0.9623	0.9951	**0.9983**

dataset, our CL-MGNet achieves the classification accuracy of 100.00% in 10 out of 15 categories and obtains the highest OA, AA, and Kappa, which is ahead of other methods.

On the Dioni dataset, our CL-MGNet achieves the highest OA of 99.32% by combining multiscale spatial information with global spectral properties, which is 0.36% higher than the second-best method, HResNetAM. Compared with transformer-based methods, the OA of our CL-MGNet is 1.45% higher than SpectralFormer and 1.03% higher than SSFTT, which indicates that the multiscale spatial information can further improve the performance of the model. Moreover, as shown in Table IX, due to the presence of hard sample (e.g., category 12) and severe sample imbalance (e.g., category 17) in the Houston2018 dataset, the classification performance is the lowest among the five datasets. For example, the classification accuracy of SpectralFormer is only 1.91% on category 12, while JigsawHSI fails completely on the classification of category 12 and category 17. However, our CL-MGNet still achieves high advantages in terms of OA, AA, and Kappa compared with other comparative methods, which indicates the superiority of simultaneously exploiting category-specific

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TABLE VII

QUANTITATIVE CLASSIFICATION RESULTS OF DIFFERENT METHODS FOR THE HOUSTON2013 DATASET (WITH 15% TRAINING SAMPLES). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Category No.	HybridSN [62]	A2S2K-ResNet [42]	SpectralNET [40]	A-SPN [63]	JigsawHSI [50]	HResNetAM [49]	SpectralFormer [45]	SSFTT [44]	Ours
1	99.25	99.92	98.76	99.96	98.87	99.85	98.68	99.81	**100.00**
2	99.91	**100.00**	99.27	99.87	**100.00**	**100.00**	98.97	99.91	99.72
3	**100.00**	**100.00**	99.93	**100.00**	**100.00**	**100.00**	99.49	**100.00**	**100.00**
4	99.34	99.92	99.15	99.91	99.81	99.99	99.72	99.91	**100.00**
5	**100.00**	**100.00**	99.77	**100.00**	**100.00**	99.89	99.72	**100.00**	**100.00**
6	96.74	**100.00**	97.03	99.57	92.75	**100.00**	96.38	**100.00**	97.83
7	95.73	99.92	96.72	99.20	97.96	99.74	96.01	98.89	**100.00**
8	98.96	**99.88**	97.98	98.09	96.59	99.76	96.59	99.53	99.72
9	95.21	98.34	98.87	98.72	98.87	99.16	94.17	98.68	**100.00**
10	**100.00**	99.18	99.87	99.87	**100.00**	99.53	98.95	**100.00**	**100.00**
11	**100.00**	99.42	99.03	**100.00**	99.90	99.77	97.81	**100.00**	**100.00**
12	**99.62**	99.41	99.39	99.58	99.52	99.29	97.04	**99.62**	**99.62**
13	97.74	**99.80**	98.35	98.80	93.48	98.54	85.96	99.25	99.75
14	**100.00**	**100.00**	99.95	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
15	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
OA(%)	98.86	99.66	98.95	99.55	98.94	99.70	97.60	99.67	**99.87**
AA(%)	98.83	99.72	98.94	99.57	98.52	99.70	97.30	99.71	**99.78**
Kappa	0.9876	0.9964	0.9886	0.9952	0.9885	0.9968	0.9741	0.9964	**0.9986**

TABLE VIII

QUANTITATIVE CLASSIFICATION RESULTS OF DIFFERENT METHODS FOR THE DIONI DATASET (WITH 10% TRAINING SAMPLES). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Category No.	HybridSN [62]	A2S2K-ResNet [42]	SpectralNET [40]	A-SPN [63]	JigsawHSI [50]	HResNetAM [49]	SpectralFormer [45]	SSFTT [44]	Ours
1	95.69	95.87	92.61	93.42	90.76	97.24	89.96	95.77	**98.42**
2	**100.00**	**100.00**	**100.00**	99.44	95.63	**100.00**	99.45	**100.00**	**100.00**
3	90.96	**98.42**	87.34	86.05	87.34	97.98	93.85	94.94	94.03
4	62.22	97.03	56.30	88.00	53.33	**100.00**	65.93	77.78	81.62
5	98.43	97.38	96.10	69.46	97.05	97.59	97.11	98.87	**99.18**
6	**100.00**	**100.00**	99.08	**100.00**	**100.00**	**100.00**	99.38	**100.00**	99.69
7	98.23	**99.78**	95.76	64.63	96.29	99.66	98.50	98.87	99.74
8	99.08	98.93	98.88	96.83	99.58	98.80	98.99	98.90	**99.63**
9	94.49	98.73	93.86	97.41	95.69	98.58	99.43	95.82	**100.00**
10	94.81	97.51	93.45	99.12	93.23	**100.00**	99.55	**100.00**	**100.00**
11	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
12	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	99.16	98.60	**100.00**
OA(%)	97.68	98.87	96.34	86.03	96.77	98.96	97.87	98.29	**99.32**
AA(%)	94.49	98.64	92.78	91.20	92.41	**99.15**	95.11	96.63	97.69
Kappa	0.9712	0.9860	0.9545	0.8256	0.9598	0.9871	0.9736	0.9787	**0.9915**

TABLE IX

QUANTITATIVE CLASSIFICATION RESULTS OF DIFFERENT METHODS FOR THE HOUSTON2018 DATASET (WITH 10% TRAINING SAMPLES). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Category No.	HybridSN [62]	A2S2K-ResNet [42]	SpectralNET [40]	A-SPN [63]	JigsawHSI [50]	HResNetAM [49]	SpectralFormer [45]	SSFTT [44]	Ours
1	85.96	81.64	**88.20**	80.94	81.55	88.03	87.01	84.05	86.13
2	95.96	97.06	95.77	96.89	95.61	95.11	94.87	94.50	**97.64**
3	**100.00**	**100.00**	**100.00**	99.84	95.62	**100.00**	98.86	**100.00**	**100.00**
4	98.47	95.75	99.32	**99.97**	98.71	96.58	97.97	97.50	99.47
5	94.09	97.09	94.59	98.54	75.86	92.97	87.23	96.72	**98.73**
6	**100.00**	99.12	**100.00**	**100.00**	**100.00**	99.45	98.62	**100.00**	**100.00**
7	99.58	80.68	94.98	**100.00**	94.56	98.65	59.83	97.49	99.58
8	97.02	97.41	99.61	98.15	95.62	96.59	92.39	98.83	**99.76**
9	99.23	99.23	99.64	96.48	99.03	99.06	98.03	99.34	**99.72**
10	90.86	95.75	93.82	80.15	91.62	90.64	76.35	92.98	**96.64**
11	85.79	92.49	89.35	62.46	79.98	88.79	70.98	90.13	**93.01**
12	35.42	**66.57**	42.26	21.84	0.00	60.91	1.91	54.09	64.60
13	95.32	93.22	98.00	91.72	93.63	94.46	86.59	96.18	**98.12**
14	97.98	96.57	99.49	99.63	89.84	94.15	88.13	96.53	**99.73**
15	99.70	98.95	99.95	99.57	**99.98**	99.05	99.14	99.70	99.94
16	98.31	95.77	99.68	99.76	93.49	95.86	94.57	98.97	**99.92**
17	76.72	**100.00**	80.17	**100.00**	0.00	**100.00**	0.00	96.55	**100.00**
18	94.23	98.70	**99.17**	93.38	98.30	96.74	78.49	95.57	**99.17**
19	98.97	98.61	**100.00**	99.25	98.56	99.46	88.63	99.40	99.95
20	99.98	**100.00**	99.56	**100.00**	99.89	99.85	98.60	99.98	**100.00**
OA	96.37	96.97	97.67	92.65	95.08	96.28	91.70	97.00	**98.40**
AA	92.18	94.23	93.68	90.93	84.09	94.32	79.91	94.43	**96.61**
Kappa	0.9522	0.9601	0.9692	0.9038	0.9350	0.9510	0.8900	0.9605	**0.9789**

properties, multiscale spatial information, and global spectral properties.

To further explore the performance of the CL-MGNet, we obtain the classification maps of our method and compared

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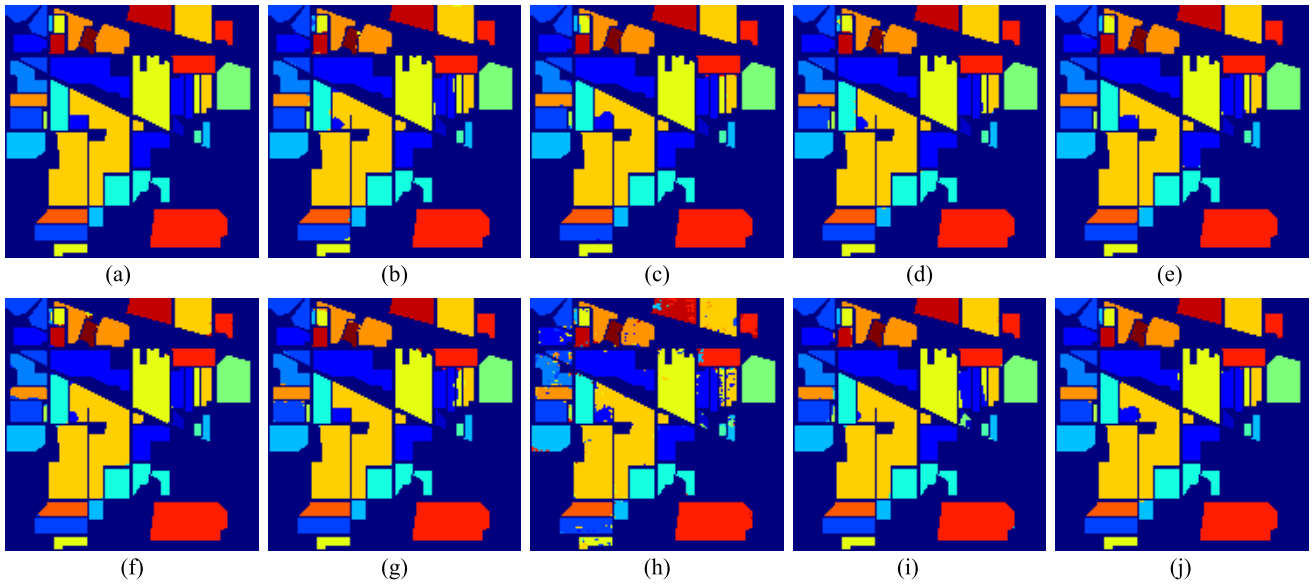


Fig. 2. Classification maps of the Indian Pines dataset. The OA results are provided in the brackets: (a) ground truth; (b) HybridSN (98.61%); (c) A2S2K-ResNet (99.37%); (d) SpectralNET (98.82%); (e) A-SPN (99.45%); (f) JigsawHSI (98.37%); (g) HResNetAM (99.25%); (h) SpectralFormer (94.20%); (i) SSFTT (97.90%); and (j) ours (99.60%).

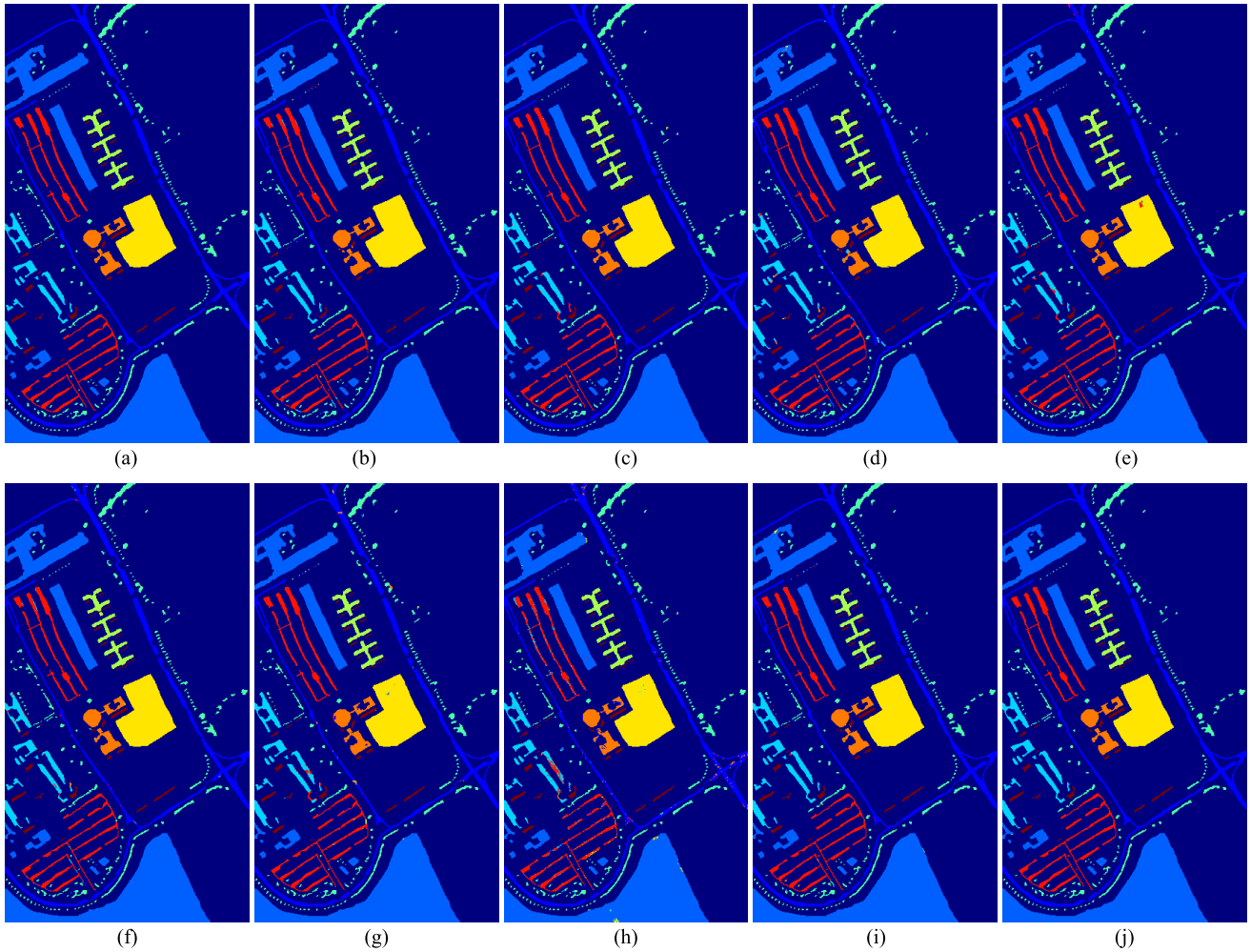


Fig. 3. Classification maps of the Pavia University dataset. The OA results are provided in the brackets: (a) ground truth; (b) HybridSN (99.39%); (c) A2S2K-ResNet (99.65%); (d) SpectralNET (99.08%); (e) A-SPN (99.51%); (f) JigsawHSI (99.29%); (g) HResNetAM (99.14%); (h) SpectralFormer (97.15%); (i) SSFTT (99.63%); and (j) ours (99.87%).

state-of-the-art methods on five hyperspectral remote sensing datasets for visual analysis. The classification maps on all five datasets are shown in Figs. 2–6, respectively. It can be seen that our method achieves near-perfect classification results on the classification of neighboring pixels in hyperspectral remote sensing images, which are quite close to the ground

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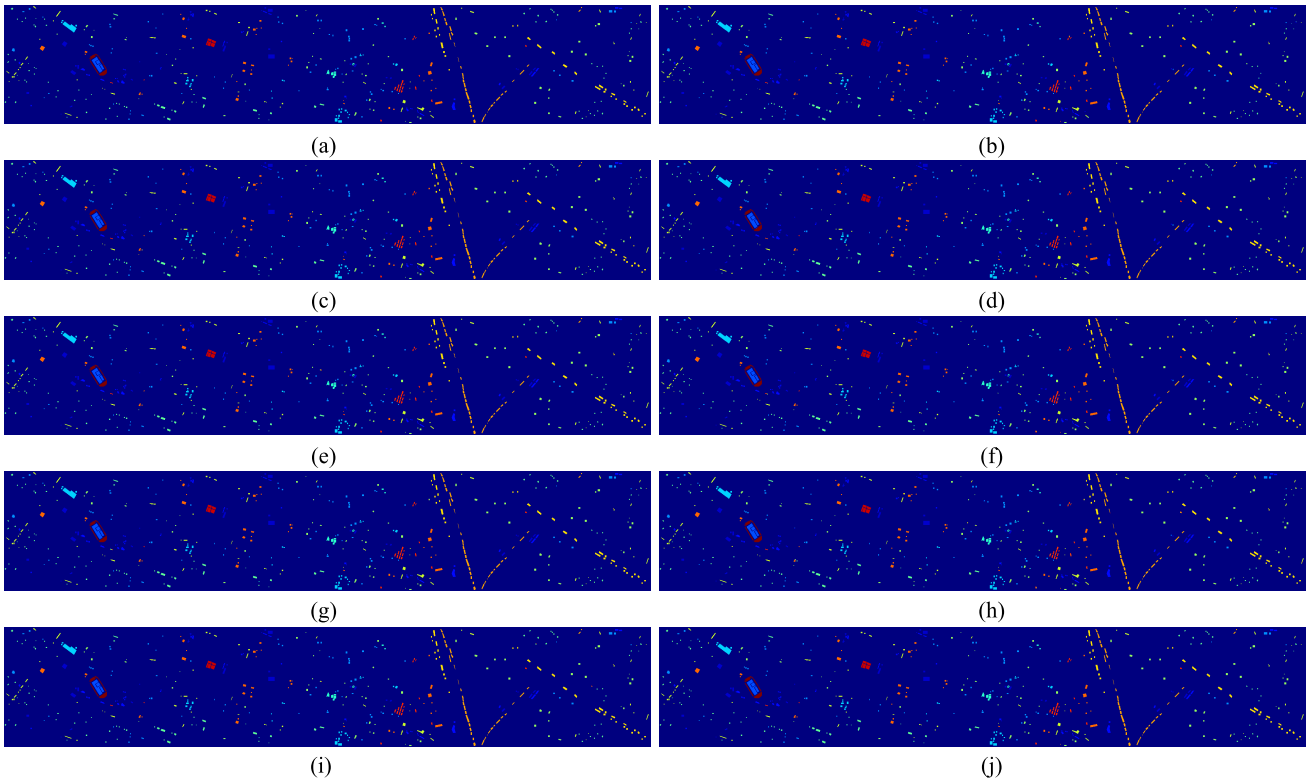


Fig. 4. Classification maps of the Houston2013 dataset. The OA results are provided in the brackets: (a) ground truth; (b) HybridSN (98.86%); (c) A2S2K-ResNet (99.66%); (d) SpectralNET (98.95%); (e) A-SPN (99.55%); (f) JigsawHSI (98.94%); (g) HResNetAM (99.70%); (h) SpectralFormer (97.60%); (i) SSFTT (99.67%); and (j) ours (99.87%).

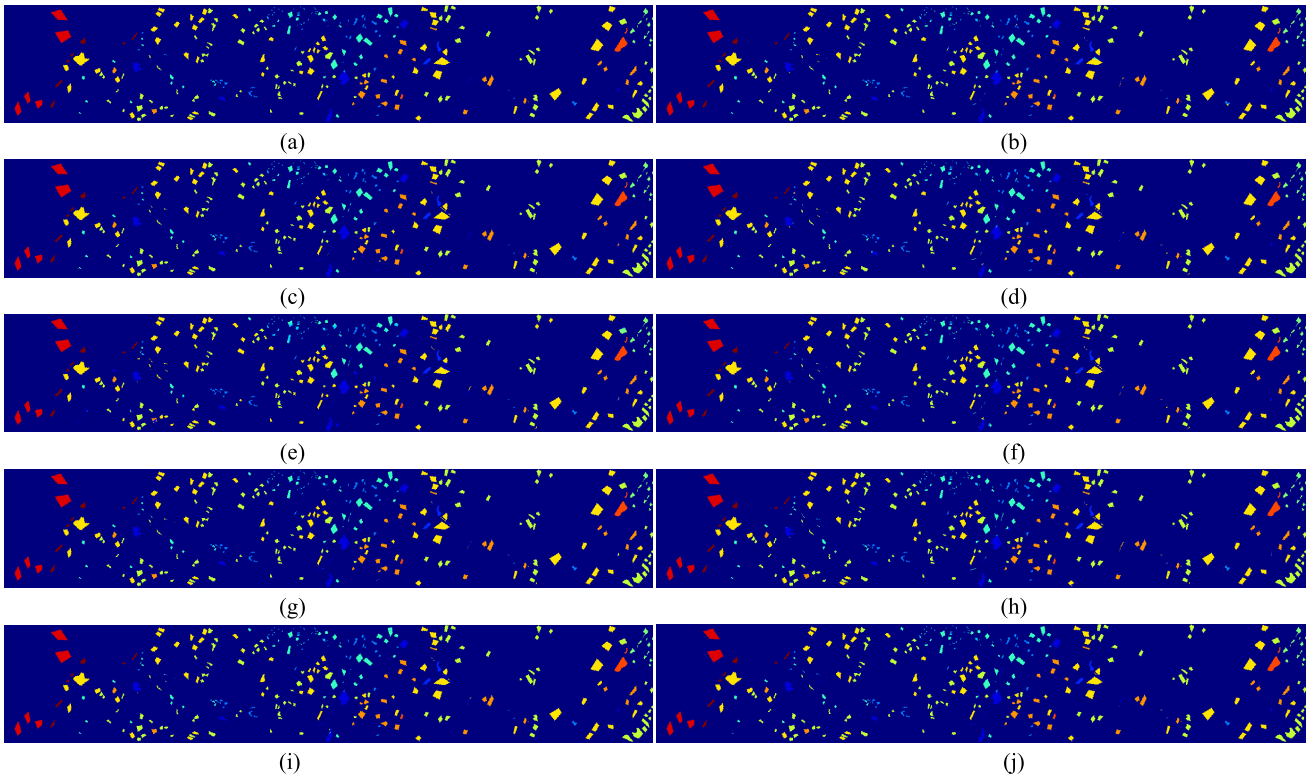


Fig. 5. Classification maps of the Dioni dataset. The OA results are provided in the brackets: (a) ground truth; (b) HybridSN (97.68%); (c) A2S2K-ResNet (98.87%); (d) SpectralNET (96.34%); (e) A-SPN (86.03%); (f) JigsawHSI (96.77%); (g) HResNetAM (98.96%); (h) SpectralFormer (97.87%); (i) SSFTT (98.29%); and (j) ours (99.32%).

truth. For example, on the Indian Pines and Houston2018 datasets, there are many misclassified neighboring pixels in the classification maps of the comparison methods. However, our CL-MGNet fully utilizes the category-specific properties

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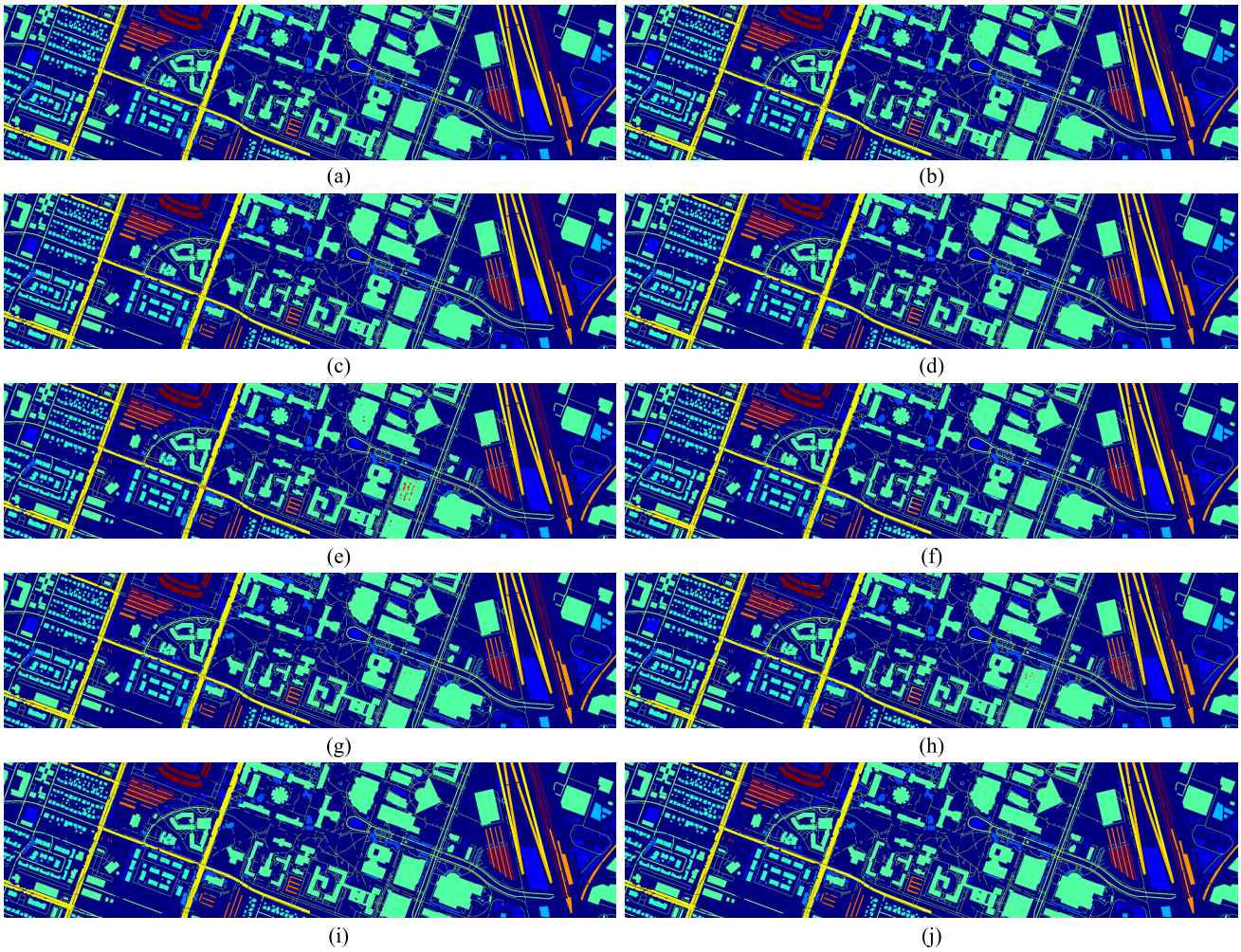


Fig. 6. Classification maps of the Houston2018 dataset. The OA results are provided in the brackets: (a) ground truth; (b) HybridSN (96.37%); (c) A2S2K-ResNet (96.97%); (d) SpectralNET (97.67%); (e) A-SPN (92.65%); (f) JigsawHSI (95.08%); (g) HResNetAM (96.28%); (h) SpectralFormer (91.70%); (i) SSFTT (97.00%); and (j) ours (98.40%).

and multiscale spatial information in hyperspectral remote sensing images, and achieves the land-cover classification of neighboring pixels well.

2) Classification Results With Spatially Disjoint Samples: To further evaluate the performance of the proposed method, we conduct experiments with spatial-disjoint training and test sets on the Indian Pines, Pavia University, Houston2013, Dioni, and Houston2018 datasets. The experimental results (in terms of OA, AA, and Kappa) are reported on Tables X and XI.

It is worth noting that there is a significant drop in classification results for spatially disjoint samples compared to randomly selected samples. There are two possible reasons for this phenomenon. For one thing, compared with spatially disjoint samples, randomly selected samples may lead to severe spatial overlapping between training and test sets, which may unfairly benefit the model and thus provide overestimated classification results. For another, in the sample selection process for some categories, the disjoint sampling strategy often results in a relatively small number of selected samples, which is challenging in terms of their complexity. Consequently, the classification results achieved using this strategy tend to be lower compared to those obtained through random sampling.

Although CL-MGNet also experiences performance degradation, it still achieves the highest OA, AA, and Kappa on five datasets. It can be seen that under the spatial-disjoint training and test sets, the improvement in classification accuracy of our method becomes more significant due to the absence of the additional performance boost provided by overlapping samples. For example, compared to the second-best method on the Indian Pines and Pavia University datasets, our method improves on OA by 3.48% and 8.24%, respectively. On the Houston2013, Dioni, and Houston2018 datasets, our method also outperforms the second-best method by 3.06%, 3.89%, and 1.85% in OA, respectively. This indicates that our method is more robust and has a greater advantage on spatially disjoint samples.

3) Space Complexity and Time Complexity: We also analyze the time complexity and space complexity of the proposed method. Taking the Indian Pines dataset as an example, the results of OA, Kappa, number of parameters, flops, and inference time of our CL-MGNet and other compared methods are listed in Table XII. Our method obtains better classification results with a reasonable level of complexity. Although there are methods (e.g., HResNetAM, SpectralFormer, and SSFTT)

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TABLE X

QUANTITATIVE CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE DISJOINT TRAINING–TEST SET FOR INDIAN PINES, PAVIA UNIVERSITY, AND HOUSTON2013 DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	Indian Pines			Pavia University			Houston2013		
	OA(%)	AA(%)	Kappa	OA(%)	AA(%)	Kappa	OA(%)	AA(%)	Kappa
HybridSN [62]	65.23	49.92	0.5995	79.97	85.12	0.7421	80.70	81.52	0.7905
A2S2K-ResNet [42]	88.64	80.81	0.8711	82.59	83.14	0.7709	87.16	89.95	0.8606
SpectralNET [40]	74.99	61.40	0.7161	85.19	81.62	0.7984	80.19	78.85	0.7848
A-SPN [63]	88.26	78.31	0.8661	84.03	91.99	0.7960	86.56	88.41	0.8540
JigsawHSI [50]	78.50	63.33	0.7550	83.10	91.04	0.7853	81.97	81.09	0.8042
HResNetAM [49]	88.68	77.88	0.8713	83.18	88.04	0.7863	88.63	90.27	0.8766
SpectralFormer [45]	83.40	73.80	0.8119	87.57	89.04	0.8354	84.26	86.28	0.8293
SSFTT [44]	83.44	77.89	0.8122	83.17	89.97	0.7845	86.75	88.94	0.8561
Ours	**91.74**	**85.15**	**0.9061**	**95.81**	**93.85**	**0.9433**	**91.69**	**91.83**	**0.9097**

TABLE XI

QUANTITATIVE CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE DISJOINT TRAINING–TEST SET FOR DIONI AND HOUSTON2018 DATASETS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	Dioni			Houston2018		
	OA(%)	AA(%)	Kappa	OA(%)	AA(%)	Kappa
HybridSN [62]	46.83	44.88	0.3652	56.74	69.81	0.4836
A2S2K-ResNet [42]	61.36	52.05	0.5211	55.76	48.91	0.4816
SpectralNET [40]	56.20	52.43	0.4744	47.00	65.95	0.3977
A-SPN [63]	55.51	57.29	0.4440	56.25	73.06	0.4894
JigsawHSI [50]	65.66	57.51	0.5751	56.41	67.28	0.4867
HResNetAM [49]	63.70	50.40	0.5563	57.48	49.34	0.4992
SpectralFormer [45]	53.71	61.26	0.4594	54.25	69.29	0.4671
SSFTT [44]	58.11	58.67	0.4775	53.36	65.89	0.4555
Ours	**69.55**	**71.77**	**0.6267**	**59.33**	**74.11**	**0.4991**

TABLE XII

SPACE COMPLEXITY AND TIME COMPLEXITY WITH DIFFERENT METHODS ON THE INDIAN PINES DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	OA (%)	Kappa	Number of Parameters	Flops	Inference Time
HybridSN [62]	98.61	0.9842	5.12M	495.84M	1.20s
A2S2K-ResNet [42]	99.37	0.9928	370.68K	103.58M	1.37s
SpectralNET [40]	98.82	0.9865	6.81M	310.85M	1.78s
A-SPN [63]	99.45	0.9937	640.98K	10.62M	0.30s
JigsawHSI [50]	98.37	0.9814	7.82M	528.58M	2.38s
HResNetAM [49]	99.25	0.9914	**65.87K**	**9.20M**	0.46s
SpectralFormer [45]	94.20	0.9339	342.65K	35.41M	1.04s
SSFTT [44]	97.90	0.9760	148.49K	11.40M	**0.21s**
Ours	**99.60**	**0.9954**	3.82M	473.65M	1.77s

with lower complexity than ours, such methods compromise accuracy and have inferior classification accuracy than ours.

F. Ablation Study

To illustrate the effectiveness of the proposed method, we take ablation experiments on the Pavia University dataset as an example. We consider six different combinations of components, and the corresponding experimental results are shown in Table XIII. Specifically, our CL-MGNet can be divided into three components, i.e., SWL module, multiscale backbone, and attention-MLP block. In case 1, we replace the EPSA Resblock in the multiscale backbone with the ResNet block [38] and explore the performance of baseline without the SWL module and the attention-MLP block. In case 2, we add the attention-MLP block to the baseline in case 1 to explore the effect of global spectral properties on the classification results. We can see that, with the attention-MLP block, this case achieves an OA of 99.73%. Then, we explore the enhancement of multiscale spatial features and category-specific information on the performance of the method, corresponding to case 3 and case 4, respectively. By using the EPSA Resblock to extract multiscale spatial features from HSI, case 3 achieves an OA of 99.84%. Meanwhile, compared with case 1, the OA of the method improves by 0.13% in the case of extracting category-specific information using the SWL module. Finally,

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TABLE XIII
ABLATION STUDY FOR THE PROPOSED CL-MGNET ON THE PAVIA UNIVERSITY DATASET (WITH 5% TRAINING SAMPLES). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Case	SWL	EPSA Resblock	Attention -MLP	Metrics		
				OA(%)	AA(%)	Kappa
1	✘	ResNet block	✘	99.67	99.30	0.9956
2	✘	ResNet block	✔	99.73	99.48	0.9965
3	✘	✔	✘	99.84	99.64	0.9979
4	✔	ResNet block	✘	99.80	99.63	0.9974
5	✔	✔	✘	99.84	99.74	0.9979
6	✔	✔	✔	**99.87**	**99.75**	**0.9983**

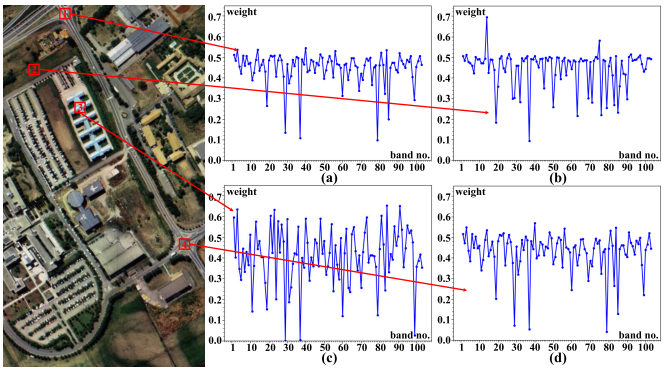


Fig. 7. Spectral weights of HSI patches belonging to different categories on the Pavia University dataset. (a) Spectral weight of HSI patch 1 belonging to category 1, (b) spectral weight of HSI patch 2 belonging to category 2, (c) spectral weight of HSI patch 3 belonging to category 5, and (d) spectral weight of HSI patch 4 belonging to category 1.

based on case 4, we use the EPSA Resblock and add the attention-MLP block sequentially to obtain the experimental results of case 5 and case 6. It can be seen that the highest OA, AA, and Kappa can be obtained by considering simultaneously the category-specific information, multiscale spatial features, and global spectral properties of HSI.

To further verify the category-specific information in the spectral bands, we use the SWL module to obtain the spectral weights of some HSI patches on the Pavia University dataset, and the results are shown in Fig. 7. As can be seen in Fig. 7(a) and (d), there is almost no difference in the spectral weights of the HSI patches belonging to the same category. However, for the HSI patches belonging to different categories, there is a large difference between their spectral weights, as shown in Fig. 7(b) and (c). The reason for this phenomenon is that different materials have different spectral reflectance characteristics. Different materials have different sensitive spectral bands, and the same spectral band has different importance for different materials.

IV. CONCLUSION

In this article, we propose a novel HSI classification method, called CL-MGNet, which can simultaneously extract category-specific properties, multiscale spatial information, and global spectral properties. The proposed CL-MGNet consists of three parts. First, we design an SWL module that uses a spectral attention mechanism to extract important spectral bands and combines it with a category consistency loss to mine the category-specific properties in spectral bands. Then, to exploit the valuable information in features at various scales while reducing the interference of redundant spatial features from less critical scales, a multiscale backbone is designed to extract the multiscale spatial feature with cross-channel attention. Next, an attention-MLP block is used to exploit the global spectral properties of HSI and obtain the feature with more discriminative information. Finally, the obtained feature is sent to a fully connected layer to obtain the land-cover category. The experimental results on five hyperspectral remote sensing datasets demonstrate the superiority of our method.

Our method shows the importance of the category-specific properties in spectral bands and the effectiveness of combining them with multiscale spatial information and global spectral properties. In our CL-MGNet, the multiscale spatial information and global spectral properties are extracted by two different modules, respectively, i.e., multiscale backbone and attention-MLP block. This combination increases the number of parameters and complexity of the model, which cannot well match the lightweight requirement. In the future, it is worth investigating how to combine multiscale convolution and global property extraction to reduce the complexity of the model while maintaining its performance.

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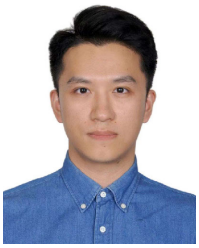
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